

MATHEMATICS DEPARTMENT

π ∞ Σ

GENERAL MATHEMATICS

MELQUI JOHN B. ARANDIA
Teacher V

Σ_{α}

Σ
 $\int$

STRENGTHENED SENIOR HIGH SCHOOL CURRICULUM



OPENING PRAYER



MATH CHECK



M

ake your self ready

A

lign your chairs properly

T

ake away all the scraps

H

ave your seats

Attendance





KAMUSTA KA?



LESSON OBJECTIVES

- Distinguish between qualitative and quantitative data, and classify real-world variables into nominal, ordinal, interval, and ratio scales.;
- Analyze data sets and their specific contexts to determine the most appropriate measure of central tendency (mean, median, mode) and variability (range, variance, standard deviation); and

LESSON OBJECTIVES

- 🎯 Formulate data-driven recommendations for a simulated barangay or school scenario by selecting correct statistical measures, demonstrating local civic awareness and integrity in data reporting.

Activity 1:

BARANGAY DATA BLITZ

Instructions:

- 1 Students receive data strips (e.g., livelihood type, captain rating, temperature in °C, monthly income, blood type, zip code).
- 2 They physically walk to designated corners: Category, Count, Rank, or Scale.
- 3 Peers validate each other's placement.



Sample Data Strips

Livelihood type:
Farming

Captain rating:
4.5 / 5

Temperature:
28°C

Monthly income:
₱15,000

Blood type:
O+

Zip code:
4000

For the Category and Rank corners, if we wanted to find the average of your data, can we add them up and divide by the number of people? Why or why not?





For the Count and Scale corners, what is the most common value in your group, and how would you find the middle value if we lined you all up from lowest to highest?

How does the way we group or measure this data affect how we summarize or analyze it?



Categorization of Statistical Data & Levels of Measurement Scale

**PRESENTED BY:
MELQUI JOHN B. ARANDIA
Teacher V**

THE DATA CARDS SORTING CHALLENGE

Instructions:

1

Step 1

Divide the class into small groups of four to five students. Each group receives a set of fifteen index cards, each containing a specific piece of information.



2

Step 2

The groups will examine the cards and organize them into different categories based on their similarities.



3

Step 3

Each group must write down their reasons for grouping the cards the way they did on a sheet of paper.



4

Step 4

Once completed, one representative from each group presents their sorted cards and explains their grouping logic to the class.



Sort • Think • Explain • Learn

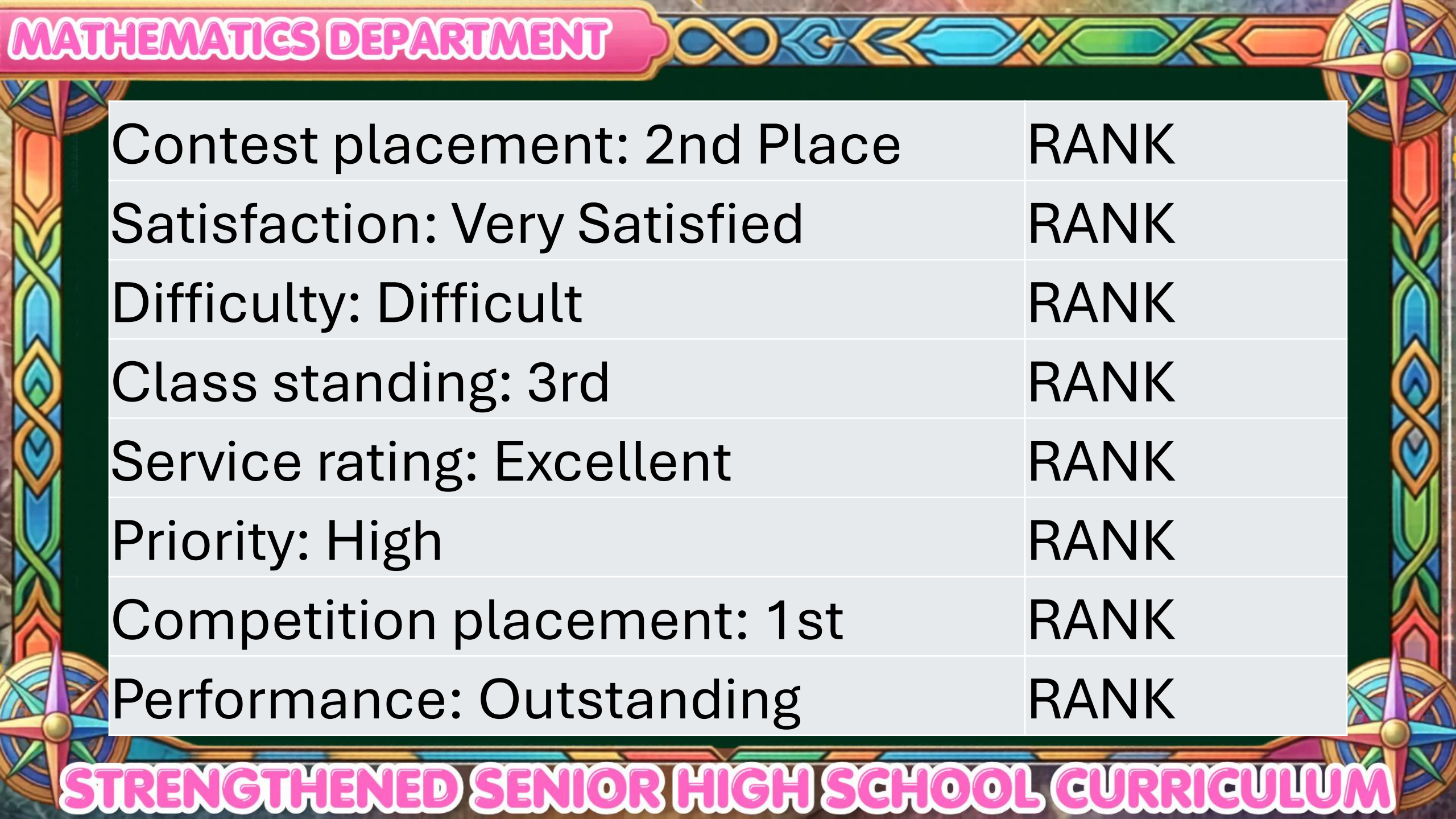
Step 1 - Divide the class into small groups of four to five students. Each group receives a set of fifteen index cards, each containing a specific piece of information.

Step 2 - The groups will examine the cards and organize them into different categories based on their similarities.

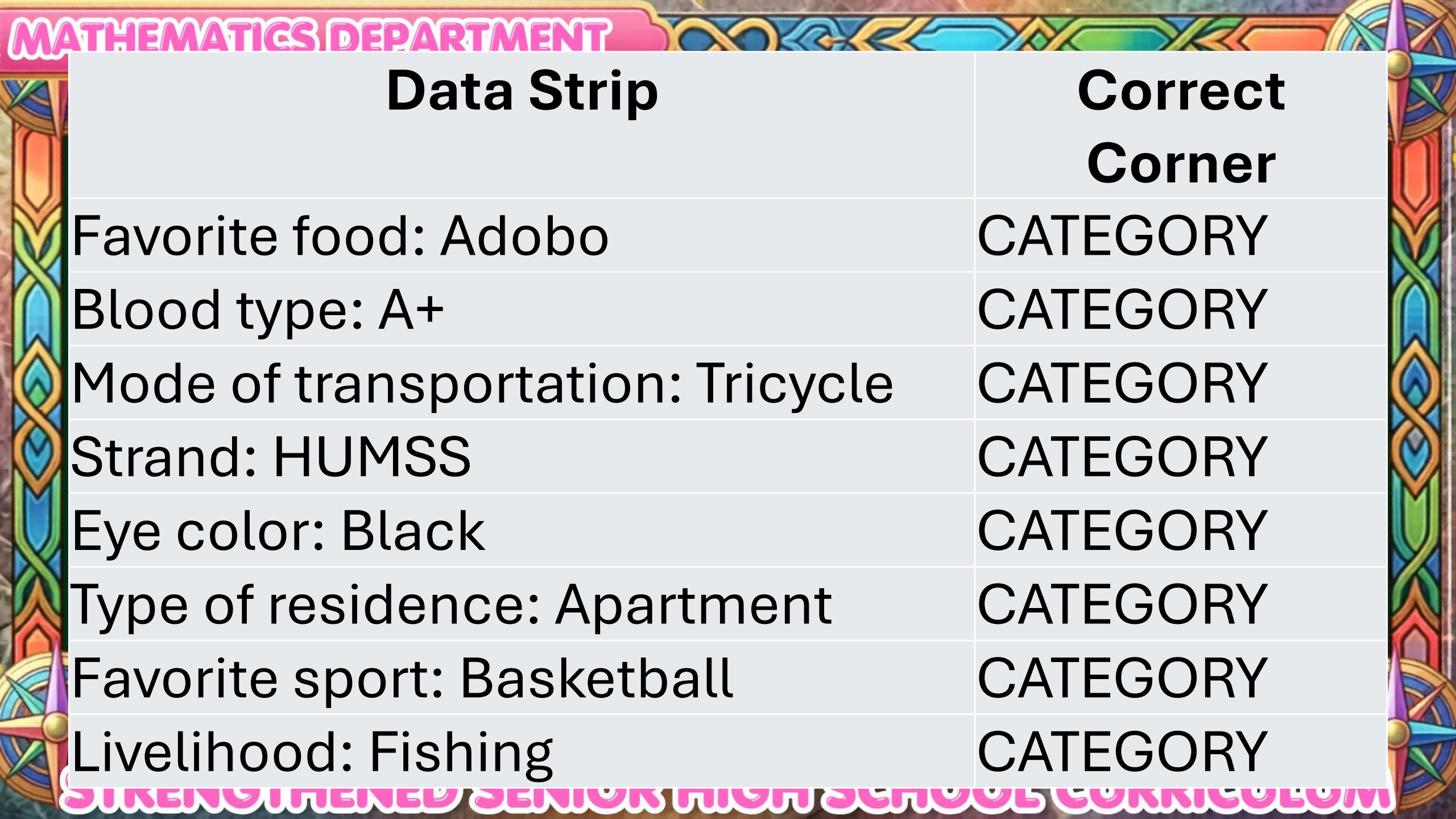
Step 3 - Each group must write down their reasons for grouping the cards the way they did on a sheet of paper.

Step 4 - Once completed, one representative from each group presents their sorted cards and explains their grouping logic to the class.

- Favorite food: Adobo
- Number of siblings: 4
- Height: 158 cm
- Contest placement: 2nd Place
- Blood type: A+
- Number of pets: 3
- Temperature: 31°C
- Satisfaction: Very Satisfied
- Mode of transportation: Tricycle
- Number of books: 15
- Monthly income: ₱20,000
- Difficulty: Difficult
- Strand: HUMSS
- Number of household members: 6
- Weight: 62.5 kg
- Class standing: 3rd
- Eye color: Black
- Number of siblings: 2
- Distance from school: 3.2 km
- Service rating: Excellent
- Type of residence: Apartment
- Number of classrooms: 14
- Age: 18 years
- Priority: High
- Favorite sport: Basketball
- Number of pets: 1
- Study time: 3.5 hours
- Competition placement: 1st
- Livelihood: Fishing
- Number of school clubs: 3
- Monthly electricity use: 210 kWh
- Performance: Outstanding



Contest placement: 2nd Place	RANK
Satisfaction: Very Satisfied	RANK
Difficulty: Difficult	RANK
Class standing: 3rd	RANK
Service rating: Excellent	RANK
Priority: High	RANK
Competition placement: 1st	RANK
Performance: Outstanding	RANK



Data Strip	Correct Corner
Favorite food: Adobo	CATEGORY
Blood type: A+	CATEGORY
Mode of transportation: Tricycle	CATEGORY
Strand: HUMSS	CATEGORY
Eye color: Black	CATEGORY
Type of residence: Apartment	CATEGORY
Favorite sport: Basketball	CATEGORY
Livelihood: Fishing	CATEGORY



Height: 158 cm

SCALE

Temperature: 31°C

SCALE

Monthly income: ₱20,000

SCALE

Weight: 62.5 kg

SCALE

Distance from school: 3.2 km

SCALE

Age: 18 years

SCALE

Study time: 3.5 hours

SCALE

Monthly electricity use: 210 kWh

SCALE



Number of siblings: 4	COUNT
Number of pets: 3	COUNT
Number of books: 15	COUNT
Number of household members: 6	COUNT
Number of siblings: 2	COUNT
Number of classrooms: 14	COUNT
Number of pets: 1	COUNT
Number of school clubs: 3	COUNT



LET'S ANALYZE

**Look at your cards.
Which groups contain
words and which
contain numbers?
What is the difference
between them?**





For the number groups, can we add or average all the numbers? Can we average room numbers or phone numbers? Why or why not?

Look at sibling ranks and canteen satisfaction. Do they have an order or ranking? How is this different from favorite colors?





How would you find the most common favorite color? Would you use the same method for exam scores or family income? Why or why not?

What happens when there are extreme values in the data, such as one family earning much more than others? How might this affect the statistical measure we use?



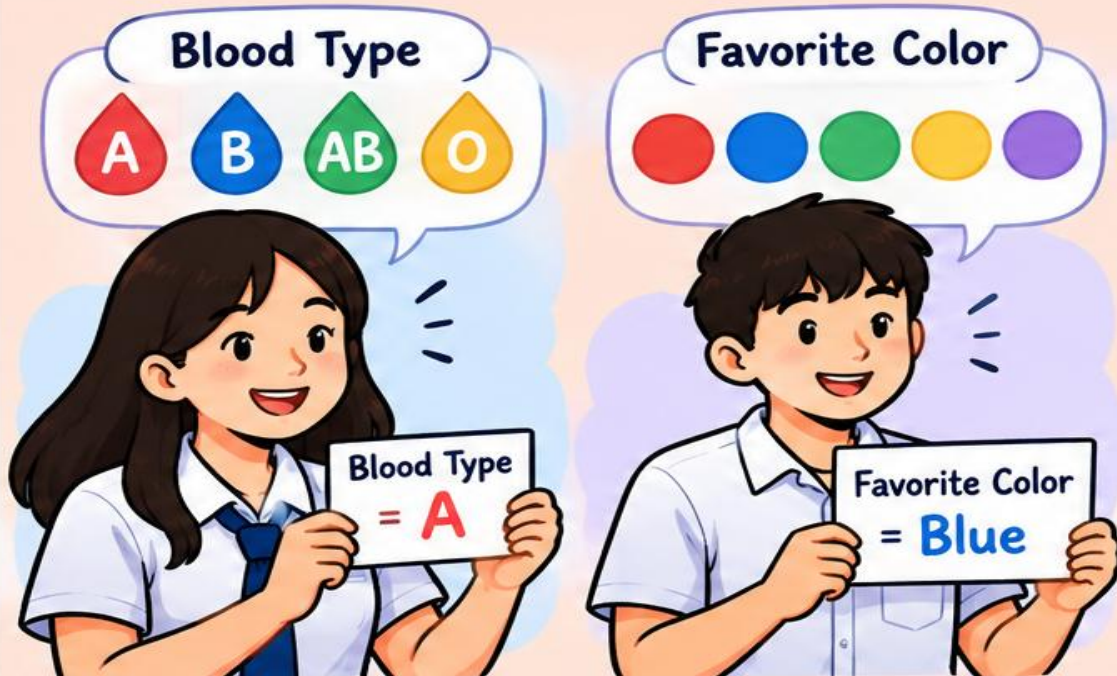


LET'S SUM IT UP!

TYPES OF DATA

QUALITATIVE OR CATEGORICAL DATA

describes attributes or characteristics.
Examples include blood type and favorite color.



QUANTITATIVE OR NUMERICAL DATA

represents counts or measurements.
This is further divided into discrete data and continuous data.

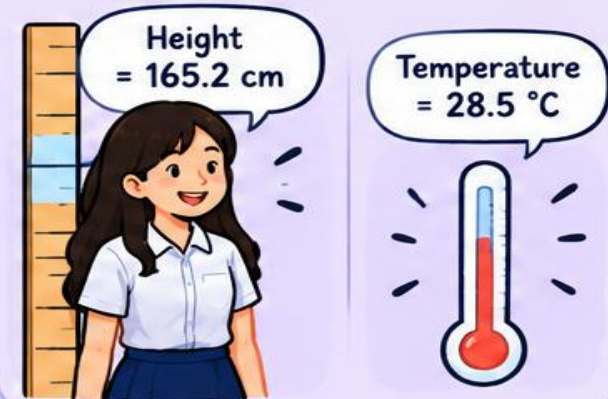
DISCRETE DATA

Countable values, like the number of siblings.



CONTINUOUS DATA

Measurable values that can take any value within a range, like height or temperature.



NOIR FRAMEWORK

A simple way to describe and organize data based on four key aspects:

NOMINAL

Names or categories

Used for labels or groups that do not have a natural order.

Gender



Favorite Color



Blood Type



Example:

Favorite color, blood type, name of barangay

ORDINAL

Ordered categories

Used for data that can be ranked or put in a specific order.



Example:

Ranking (1st, 2nd, 3rd), level of satisfaction (low, medium, high)

INTERVAL

Equal intervals, no true zero

Used for data with equal differences between values, but the zero point does not mean "none".



Temperature (°C)

Example:

Temperature (°C), yearly income (in thousands)

RATIO

True zero, meaningful ratios

Used for data with equal intervals and a true zero point. Ratios between values are meaningful.



Height (cm)
165 cm

Weight (kg)
60 kg



Example:

Height, weight, number of siblings, monthly income

Appropriate Measures of Central Tendency and Variability Based on Data Type and Purpose

The right measure depends on the type of data and what you want to find out.

NOMINAL DATA

Categories or names. There is no natural order.

Example:

Favorite color of students



Red



Blue



Green



Yellow

Central Tendency

- Mode – the most frequent category.

Variability

- Does not apply to nominal data.



Example: Mode = Blue
(if it appears most often)

Appropriate Measures of Central Tendency and Variability Based on Data Type and Purpose

The right measure depends on the type of data and what you want to find out.

ORDINAL DATA

Categories with a natural order, but the differences between categories are not equal.

Example:

Class rank in a competition



Central Tendency

- Median – the middle rank.
- Mode – can also be used.

Variability

- Range of ranks (highest rank – lowest rank).



Example: Median = 3rd place
(Mode = 1st place if most common)

Appropriate Measures of Central Tendency and Variability Based on Data Type and Purpose

The right measure depends on the type of data and what you want to find out.

INTERVAL DATA

Equal intervals, no true zero.
Ratios are not meaningful.

Example:

Temperature ($^{\circ}\text{C}$)



30°C
25°C
20°C
15°C
10°C



Example: Mean temperature = 20°C
(If there are extreme values, use median instead.)

Central Tendency

- **Mean** – preferred (uses all data points).
- **Median** – if there are extreme outliers (not affected by extreme values).

Variability

- **Range** – quick estimate of spread.
- **Standard Deviation** – most reliable (shows how far data points deviate from the mean).

Appropriate Measures of Central Tendency and Variability Based on Data Type and Purpose

The right measure depends on the type of data and what you want to find out.

RATIO DATA

True zero, meaningful ratios.

Example:

Height of students (cm)



Example: Mean height = 162.5 cm
(If there are outliers, use median.)

Central Tendency

- **Mean** – preferred (uses all data points).
- **Median** – if there are extreme outliers (not affected by extreme values).

Variability

- **Range** – quick estimate of spread.
- **Standard Deviation** – most reliable (shows how far data points deviate from the mean).

Appropriate Measures of Central Tendency and Variability Based on Data Type and Purpose

The right measure depends on the type of data and what you want to find out.

NOMINAL DATA

Categories or names. There is no natural order.

Example:

Favorite color of students



Central Tendency

- Mode – the most frequent category.

Variability

- Does not apply to nominal data.



Example: Mode = Blue (if it appears most often)

ORDINAL DATA

Categories with a natural order, but the differences between categories are not equal.

Example:

Class rank in a competition



Central Tendency

- Median – the middle rank.
- Mode – can also be used.

Variability

- Range of ranks (highest rank – lowest rank).



Example: Median = 3rd place (Mode = 1st place if most common)

INTERVAL DATA

Equal intervals, no true zero. Ratios are not meaningful.

Example:

Temperature (°C)



Central Tendency

- Mean – preferred (uses all data points).
- Median – if there are extreme outliers (not affected by extreme values).

Variability

- Range – quick estimate of spread.
- Standard Deviation – most reliable (shows how far data points deviate from the mean).



Example: Mean temperature = 20°C (If there are extreme values, use median instead.)

RATIO DATA

True zero, meaningful ratios.

Example:

Height of students (cm)



Central Tendency

- Mean – preferred (uses all data points).
- Median – if there are extreme outliers (not affected by extreme values).

Variability

- Range – quick estimate of spread.
- Standard Deviation – most reliable (shows how far data points deviate from the mean).



Example: Mean height = 162.5 cm (If there are outliers, use median.)



**LET'S
APPLY**

Activity 3: THE COMMUNITY HEALTH AND NUTRITION ADVOCACY TASK

Scenario:

The Barangay Health Center has collected various health metrics from fifty resident children to design a new nutrition program. The collected data includes:



Activity 3: THE COMMUNITY HEALTH AND NUTRITION ADVOCACY TASK

1. Household ID number.
2. Nutritional status classification (Severely Underweight, Underweight, Normal, Overweight).
3. Exact weight of each child in kilograms.
4. Body temperature in Celsius taken during the check-up.
5. Daily vegetable intake measured in number of servings.



With your group, complete the following tasks on a worksheet:

Task A: Identify the type of data (qualitative or quantitative) and the level of measurement (nominal, ordinal, interval, or ratio) for each of the five-health metrics collected.

Task B: For each metric, determine the most appropriate measure of central tendency (mean, median, or mode) and measure of variability (range or standard deviation) that the barangay health workers should use to summarize the community's health profile. Write a short, two-sentence justification for each of your choices.

Task C: Imagine the data shows that most children eat zero to one serving of vegetables daily, but one child eats ten servings. Explain which measure of central tendency the health center should use to report the typical vegetable intake of the children and explain why.



MATHEMATICS DEPARTMENT

EXIT TICKET

STRENGTHENED SENIOR HIGH SCHOOL CURRICULUM

⇒ EXIT TICKET

1. Which of the following is an example of qualitative data?

- A. Age of students**
- B. Number of siblings**
- C. Favorite color**
- D. Exam score**



⇒ EXIT TICKET

2. A teacher records students' satisfaction as "Very Satisfied, Satisfied, Neutral, Dissatisfied." What level of measurement is this?

A. Nominal

C. Interval

B. Ordinal

D. Ratio



⇒ EXIT TICKET

3. Which level of measurement is used for data with categories that have no order?

☒ A. Nominal

C. Interval

B. Ordinal

D. Ratio



⇒ EXIT TICKET

4. What statistical measure represents the middle value when data are arranged from lowest to highest?

- A. Mean**
- B. Mode**

- ☒ C. Median**
- D. Range**



⇒ EXIT TICKET

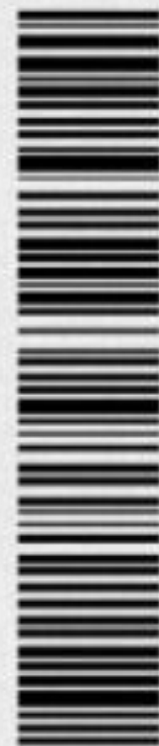
5. A student's scores are 80, 85, 90, 95, and 100. What is the mean score?

A. 85

B. 88

C. 90

D. 95



WAYS FORWARD

Consumer Report Critique

Students will select a print, digital, or social media advertisement or a news article that uses statistics to promote a product or report a trend. They will analyze the claim being made. In a short, three-paragraph reflection, they will:

WAYS FORWARD

- 1. Identify the variables used in the report and classify their levels of measurement.**
- 2. Evaluate whether the advertisers or reporters used the correct measures of central tendency and variability. For example, if a company claims the average salary of its employees is high, did they use the mean to hide extreme outliers, when the median would have been more accurate?**
- 3. Suggest a more mathematically sound and ethical way to present the data.**

SESSION DEVELOPER

JOANA MARIE V. ANCIADO
Teacher II



REFERENCE(S):

- De Guzman, D. B., & Quintana, C. A. (2017). *General mathematics for senior high school*. C & E Publishing, Inc., pages 126-132
- <https://www.ga.gov.au/scientific-topics/national-location-information/topographic-maps-data/basics/what-is-map-scale>
- https://www.mathwords.com/s/scale.htm?utm_source
- https://education.nationalgeographic.org/resource/map-scale/?utm_source
- https://www.archives.gov/research/catalog/lcdrg/elements/scale-note?utm_source